SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: CSA51
COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Analytical Problem Solving: 40%

QUESTIONS: Total Marks: 50 Annexure A

Code of Conduct:

I _BANAGANAPALLI MUNNA(192372028)__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: B.Munna

Annexure A
Analytical Problem Solving

1. RSA Encryption Analysis
Given p=11, q=13

a. Compute n ϕ(n)\
b. Choose e=7 and find the private key d
c. Encrypt the message M=9
d. Decrypt the ciphertext and verify the result
e. Analyze how changing e affects security

2. Block Cipher Mode Evaluation
Given plaintext blocks:

P=[1010,1010,1111,1010], Key = K=1100

a. Encrypt using ECB and CBC modes (Assume IV = 0000)
b. Compare the ciphertext outputs
c. Analyze which mode hides patterns better and explain why

3. DES vs 3-DES vs Blowfish Performance Study
Given:

a. DES key size = 56 bits, time = 2 ms/block
b. 3-DES key size = 168 bits, time = 6 ms/block
c. Blowfish key size = 128 bits, time = 1 ms/block

For 1000 blocks of data:

i.Calculate total encryption time for each algorithm
ii.Analyze trade-offs between security and performance
iii.Recommend the best algorithm for secure real-time communication

4. Diffie-Hellman Key Exchange Analysis
Given:

a. p=23, g=5
b. User A private key = 6
c. User B private key = 15
d. Compute public keys for A and B
e. Compute the shared secret key
f. Demonstrate how a Man-in-the-Middle attacker could alter the exchange
g. Suggest a secure method to prevent the attack

5. ECC vs RSA Comparison
Given:

a. RSA key size = 1024 bits
b. ECC key size = 160 bits (equivalent security)
c. Device processing capacity = limited (IoT device)
d. Analyze computational cost for both systems
e. Estimate which consumes less power and memory
f. Justify which cryptosystem is better for the given scenario with reasoning

. RSA Encryption Analysis

Given:

- p=11p = 11p=11
- q=13q = 13q=13

a. Compute nnn and ϕ(n)\phi(n)ϕ(n)

1. Compute nnn:

   n=p×q=11×13=143n = p \times q = 11 \times 13 = 143n=p×q=11×13=143

2. Compute Euler's Totient:

   ϕ(n)=(p−1)(q−1)\phi(n) = (p-1)(q-1)ϕ(n)=(p−1)(q−1) =(11−1)(13−1)=10×12=120= (11-1)(13-1) = 10 \times 12 = 120=(11−1)(13−1)=10×12=120

Answer:

- n=143n = 143n=143
- ϕ(n)=120\phi(n) = 120ϕ(n)=120

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b. Choose e=7e = 7e=7 and find the private key ddd

We need ddd such that:

d×e≡1(mod120)d \times e \equiv 1 \pmod{120}d×e≡1(mod120)

So,

7d≡1(mod120)7d \equiv 1 \pmod{120}7d≡1(mod120)

Using the Extended Euclidean Algorithm:

7×103=7217 \times 103 = 7217×103=721 721mod 120=1721 \mod 120 = 1721mod120=1

Therefore,

d=103d = 103d=103

Private Key:

- Public key = (7, 143)

- Private key = (103, 143)

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c. Encrypt the message M=9M = 9M=9

Encryption formula:

C=Me mod nC = M^e \bmod nC=Memodn C=97 mod 143C = 9^7 \bmod 143C=97mod143

Compute:

- 92=819^2 = 8192=81
- 94=812=6561≡126(mod143)9^4 = 81^2 = 6561 \equiv 126 \pmod{143}94=812=6561≡126(mod143)
- 97=94×92×99^7 = 9^4 \times 9^2 \times 997=94×92×9

126×81×9≡48(mod143)126 \times 81 \times 9 \equiv 48 \pmod{143}126×81×9≡48(mod143)

Ciphertext:

C=48\boxed{C = 48}C=48

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d. Decrypt the ciphertext

Decryption formula:

M=Cd mod nM = C^d \bmod nM=Cdmodn M=48103 mod 143M = 48^{103} \bmod 143M=48103mod143

Using modular exponentiation,

48103 mod 143=948^{103} \bmod 143 = 948103mod143=9

Recovered message:

M=9\boxed{M = 9}M=9

The decrypted message matches the original, verifying the RSA process.

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e. Analyze how changing eee affects security

- A **small value of eee** (such as 3 or 5) makes encryption faster but can introduce vulnerabilities if padding is not used correctly.
- A **moderate value**, such as **65537**, is widely used because it offers efficient encryption and strong security.

- A **large value of eee** increases encryption time but provides little additional practical security.
- The value of eee must satisfy:
  - 1<e<ϕ(n)1 < e < \phi(n)1<e<ϕ(n)
  - gcd(e,ϕ(n))=1\gcd(e,\phi(n)) = 1gcd(e,ϕ(n))=1

Conclusion: Choosing an appropriate public exponent (typically 65537) balances performance and security.

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5. ECC vs RSA Comparison

Given

- RSA key size = **1024 bits**
- ECC key size = **160 bits**
- IoT device with limited processing power

Analyze computational cost

Feature	RSA (1024-bit)	ECC (160-bit)
Key size	Large	Small
Computation	Large integer arithmetic	Elliptic curve arithmetic
Encryption	Fast	Moderate
Decryption/Signing	Computationally expensive	More efficient
Overall processing	Higher CPU usage	Lower CPU usage

Interpretation

ECC uses much smaller keys while providing equivalent security, resulting in lower computational cost and faster execution on resource-constrained devices.

(e) Estimate power and memory consumption

Resource	RSA	ECC
Memory usage	High	Low
Storage requirement	High	Low
CPU utilization	High	Low
Battery consumption	High	Low
Communication bandwidth	High	Low

Interpretation

ECC consumes significantly **less memory, storage, processing power, and battery energy** than RSA, making it ideal for IoT devices.

f) Recommend the better cryptosystem

Recommended: Elliptic Curve Cryptography (ECC)

Reasons

1. Provides equivalent security with a much smaller key (160-bit ECC ≈ 1024-bit RSA).
2. Requires less RAM and storage.
3. Performs cryptographic operations more efficiently on low-power processors.
4. Reduces battery consumption, increasing IoT device lifetime.
5. Smaller keys and signatures reduce communication bandwidth.
6. Better suited for embedded and real-time applications.

Final Conclusion

For a **low-power IoT device**, **ECC** is the preferred cryptosystem because it delivers the same security as RSA while requiring **less computation, memory, storage, bandwidth, and power**. This makes ECC the most efficient and practical choice for secure communication in smart home and IoT environments.

Summary

RSA Results

- n=143n = 143n=143
- ϕ(n)=120\phi(n) = 120ϕ(n)=120
- Public key = (7,143)(7,143)(7,143)

- Private key = (103,143)(103,143)(103,143)
- Plaintext = **9**
- Ciphertext = **48**
- Decrypted message = **9** (Verified)

ECC vs RSA

- **RSA (1024-bit):** Higher computation, memory, and power consumption.
- **ECC (160-bit):** Equivalent security with lower computation, memory usage, bandwidth, and power consumption.
- **Best choice for IoT: ECC**.

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method

strong justification

Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorgani ze d steps; multiple errors	No clear process
Accuracy of Calculatio n	20%	All calculatio ns accurate; correct final answer	Minor calculati on errors	Frequent errors; incorrect final answer	No valid calculati on s
Interpreta tio n of Results	15%	Clearly interprets results with units and engineeri ng relevance	Basic interpret ati on with minor omissions	Limited or unclear interpreta tio n	No interpret ati on